

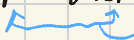
Last Time:

- Course Logistics
- History of Dynamics
- (Started) Newtonian Mechanics Review

Today:

- Newtonian Review Pt. 2
- Much ado about Coordinates
- Three views of the Pendulum

Newtonian Mechanics (From last time):

- Motion of particles / point masses
- $\vec{F} = m\vec{a}$ only holds when:
 -  abstract vectors (coordinate free)
 - * Mass is constant
 - * Vectors are expressed in an "inertial" / "Newtonian" reference frame (not accelerating or rotating)
- $\vec{F} = m\vec{a}$ (in components) only holds when coordinates are Cartesian
- Roughly: reference frame $\sim$ rigid body. Has an origin. Can accelerate / rotate. Doesn't define coordinates.
- Coordinate system $\sim$ grid we use to assign numbers (components) to vectors.

-
- We can change coordinates without changing the underlying vector:

$$\vec{v} = {}^e v^i \hat{e}_i = {}^d v^j \hat{d}_j$$

$$= [\hat{e}_1 \hat{e}_2 \hat{e}_3] \begin{bmatrix} e_{v_1} \\ e_{v_2} \\ e_{v_3} \end{bmatrix} = [\hat{d}_1 \hat{d}_2 \hat{d}_3] \begin{bmatrix} d_{v_1} \\ d_{v_2} \\ d_{v_3} \end{bmatrix}$$

* Multiply both sides $\hat{d}$:

$$\Rightarrow \hat{d} \hat{e}^T e_v = \hat{d} \hat{d}^T d_v$$

$$\hat{d} \hat{d}^T = \begin{bmatrix} \hat{d}_1 \\ \hat{d}_2 \\ \hat{d}_3 \end{bmatrix} [\hat{d}_1 \hat{d}_2 \hat{d}_3] = \begin{bmatrix} \hat{d}_1 \cdot \hat{d}_1 & \hat{d}_1 \cdot \hat{d}_2 & \hat{d}_1 \cdot \hat{d}_3 \\ \hat{d}_2 \cdot \hat{d}_1 & \hat{d}_2 \cdot \hat{d}_2 & \hat{d}_2 \cdot \hat{d}_3 \\ \hat{d}_3 \cdot \hat{d}_1 & \hat{d}_3 \cdot \hat{d}_2 & \hat{d}_3 \cdot \hat{d}_3 \end{bmatrix} = I \quad (\text{if basis is orthogonal})$$

$$\Rightarrow [\hat{d} \hat{e}^T] e_v = d_v$$

$$\begin{bmatrix} \hat{d}_1 \\ \hat{d}_2 \\ \hat{d}_3 \end{bmatrix} [\hat{e}_1 \hat{e}_2 \hat{e}_3] = \begin{bmatrix} \hat{d}_1 \cdot \hat{e}_1 & \hat{d}_1 \cdot \hat{e}_2 & \hat{d}_1 \cdot \hat{e}_3 \\ \hat{d}_2 \cdot \hat{e}_1 & \hat{d}_2 \cdot \hat{e}_2 & \hat{d}_2 \cdot \hat{e}_3 \\ \hat{d}_3 \cdot \hat{e}_1 & \hat{d}_3 \cdot \hat{e}_2 & \hat{d}_3 \cdot \hat{e}_3 \end{bmatrix}$$

${}^d R^e$

(rotation from e to d coordinates)

* Note $e_v = [\hat{e} \hat{d}^T] d_v = {}^e R^d d_v$

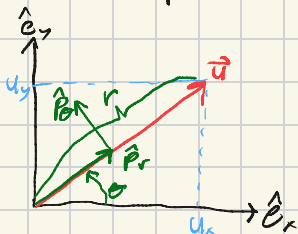
$$\Rightarrow {}^e R^d = ({}^d R^e)^T = ({}^d R^e)^{-1}$$

* Interpretation:

- Rows of ${}^d R^e$ are $\hat{d}$ basis vectors expressed in $\hat{e}$ components
- Columns are $\hat{e}$ basis vectors expressed in $\hat{d}$ components

$${}^d R^e = \begin{bmatrix} {}^e d_1^T \\ {}^e d_2^T \\ {}^e d_3^T \end{bmatrix} = [{}^d e_1 \quad {}^d e_2 \quad {}^d e_3]$$

- We also need to deal with non-Cartesian ("curvilinear") coordinates.
- A classic example - Polar coordinates:



$$\begin{aligned} \vec{v} &= u_x \hat{e}_x + u_y \hat{e}_y \\ &= r \hat{p}_r \end{aligned}$$

* Unlike the Cartesian case, change of coordinates is nonlinear:

$$u_x = r \cos(\theta) \quad , \quad u_y = r \sin(\theta)$$

$$r = \sqrt{u_x^2 + u_y^2} \quad , \quad \theta = \text{atan2}(u_y, u_x)$$

$$(\text{= atan}(u_y/u_x) \text{ if } u_x, u_y \geq 0)$$

* Also unlike Cartesian coordinates, we don't have constant basis vectors:

$$\left. \begin{aligned} \hat{p}_r &= \cos(\theta) \hat{e}_x + \sin(\theta) \hat{e}_y \\ \hat{p}_\theta &= -\sin(\theta) \hat{e}_x + \cos(\theta) \hat{e}_y \end{aligned} \right\} = \begin{bmatrix} \hat{p}_r \\ \hat{p}_\theta \end{bmatrix} = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} \hat{e}_x \\ \hat{e}_y \end{bmatrix}$$

* Note: we don't write position $\vec{u} = u_r \hat{p}_r + u_\theta \hat{p}_\theta$ like we did in Cartesian coordinates (but we can do this for the velocity + acceleration).

- ▶ The $\hat{p}$ basis vectors are better understood as pointing in the direction of incremental increase in each coordinate (think $\sim$ gradient). They are useful for computing derivatives (velocity, acceleration).

- Velocity in polar coordinates:

$$\begin{aligned}\vec{u} &= \dot{u}_x \hat{e}_x + \cancel{\dot{u}_x \hat{e}_x} + \dot{u}_y \hat{e}_y + \cancel{\dot{u}_y \hat{e}_y} = \dot{u}_x \hat{e}_x + \dot{u}_y \hat{e}_y \\ &= \dot{r} \hat{p}_r + r \dot{\theta} \hat{p}_\theta\end{aligned}$$

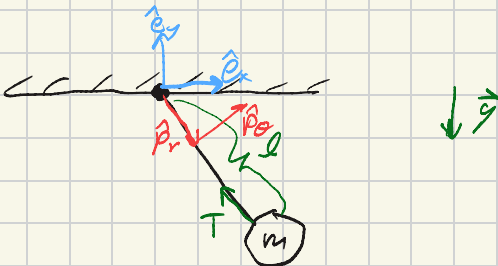
$$\hat{p}_r = -\sin(\theta) \hat{e}_x + \cos(\theta) \hat{e}_y = \hat{e}_\theta$$

$$\hat{p}_\theta = -\cos(\theta) \hat{e}_x - \sin(\theta) \hat{e}_y = -\hat{e}_r$$

$$\Rightarrow \vec{u} = \dot{r} \hat{p}_r + r \dot{\theta} \hat{p}_\theta$$

$$\begin{aligned}\Rightarrow \vec{a} &= \ddot{r} \hat{p}_r + \dot{r} \dot{\hat{p}}_r + \dot{r} \dot{\theta} \hat{p}_\theta + r \ddot{\theta} \hat{p}_\theta + r \dot{\theta} \dot{\hat{p}}_\theta \\ &= \dot{r} \dot{\hat{p}}_r + \dot{r} \dot{\theta} \hat{p}_\theta + r \ddot{\theta} \hat{p}_\theta - r \dot{\theta}^2 \hat{p}_r \\ &= (\ddot{r} - r \dot{\theta}^2) \hat{p}_r + (r \ddot{\theta} + 2\dot{r} \dot{\theta}) \hat{p}_\theta\end{aligned}$$

Example: Pendulum



- Planar/2D system has 1 constraint $\Rightarrow$ 1 DOF
 $r = \|\vec{r}(t)\| = l$

- Newton's 2nd Law:

$$\vec{F} = \vec{T} + m\vec{g} = m\vec{\ddot{r}} \quad \leftarrow \text{"coordinate free"}$$

- We need to choose coordinates to compute anything:

$$\vec{F} = \begin{bmatrix} -T \\ 0 \end{bmatrix}^T \begin{bmatrix} \hat{p}_r \\ \hat{p}_\theta \end{bmatrix} + \begin{bmatrix} 0 \\ -mg \end{bmatrix}^T \begin{bmatrix} \hat{e}_x \\ \hat{e}_y \end{bmatrix} = m \begin{bmatrix} \ddot{r}_x \\ \ddot{r}_y \end{bmatrix}^T \begin{bmatrix} \hat{e}_x \\ \hat{e}_y \end{bmatrix} = m \begin{bmatrix} \ddot{r} - r\dot{\theta}^2 \\ r\ddot{\theta} + 2\dot{r}\dot{\theta} \end{bmatrix}^T \begin{bmatrix} \hat{p}_r \\ \hat{p}_\theta \end{bmatrix}$$

- Playbook: put everything in the same coordinates so we can "factor them out" to get ODEs for the coordinates.

* Polar:

$$\hat{e}_y = \sin(\theta) \hat{p}_r + \cos(\theta) \hat{p}_\theta$$

$$\left(\begin{bmatrix} -T \\ 0 \end{bmatrix} + \begin{bmatrix} mgl \sin(\theta) \\ -mgl \cos(\theta) \end{bmatrix} - m \begin{bmatrix} \ddot{r} - r\dot{\theta}^2 \\ r\ddot{\theta} + 2\dot{r}\dot{\theta} \end{bmatrix} \right)^T \begin{bmatrix} \hat{p}_r \\ \hat{p}_\theta \end{bmatrix} = 0$$

$$\begin{cases} m(\ddot{r} - r\dot{\theta}^2) + T + mgl \sin(\theta) = 0 \\ m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) + mgl \cos(\theta) = 0 \end{cases}$$

* Note: $r = l \Rightarrow \dot{r} = \ddot{r} = 0$

$$\Rightarrow \begin{cases} -ml\dot{\theta}^2 + T + mgl \sin(\theta) = 0 & (1) \\ ml\ddot{\theta} + mgl \cos(\theta) = 0 & (2) \end{cases}$$

- (2) is actually all we need to simulate the system

- (1) gives us the constraint force T if we want

$$\Rightarrow T = m l \dot{\theta}^2 - m g l \sin(\theta) = \frac{m(\dot{\mathbf{r}}^T \dot{\mathbf{r}})}{l} - m g r_y$$

(if we don't we don't want T , we can ignore (1))

$\Rightarrow$ Careful choice of coordinates can make life easier.

* Cartesian:

$$\hat{\mathbf{p}}_r = \cos(\theta) \hat{\mathbf{e}}_x + \sin(\theta) \hat{\mathbf{e}}_y = \frac{r_x}{l} \hat{\mathbf{e}}_x + \frac{r_y}{l} \hat{\mathbf{e}}_y$$

$$\Rightarrow \left(\begin{bmatrix} -T r_x / l \\ -T r_y / l \end{bmatrix} + \begin{bmatrix} 0 \\ -m g \end{bmatrix} - m \begin{bmatrix} \ddot{r}_x \\ \ddot{r}_y \end{bmatrix} \right)^T \begin{bmatrix} \hat{\mathbf{e}}_x \\ \hat{\mathbf{e}}_y \end{bmatrix} = 0$$

$$\begin{cases} m \ddot{r}_x + \frac{T}{l} r_x = 0 \\ m \ddot{r}_y + \frac{T}{l} r_y + m g = 0 \end{cases}$$

* Note: T explicitly shows up in the ODE here, so we have to compute it. We also have to deal with 2 ODEs instead of 1.

* Pendulum Simulation:

- Cartesian version exhibits constraint drift due to numerical integration error.

$\Rightarrow$ Need additional constraint stabilization (PD feedback). Called "Baumgarte Stabilization".